Liouville geometry and the full skeleton

A geometric proof for unimodular acyclic cluster models

Cluster skeleton formalization project

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Abstract

We construct a smooth exact symplectic structure on the full exchange variety of an integral unimodular skew-symmetric seed whose mutable quiver is acyclic. Frozen coordinates are retained as invertible variables. The construction extends the imaginary part of the logarithmic two-form from the initial torus, has a Liouville vector field complete in both time directions, and identifies its full trapped set with an explicit compact coordinate locus. The proof treats all simultaneous-zero strata and uses the same vector field and complete flow throughout. This note gives the mathematical argument for the geometric statement formalized in the accompanying Lean project.

1 The exchange variety and the precise statement

Let $E = \{1, \dots, d\}$, with a partition $E = M \sqcup F$ into mutable and frozen indices. Let $B = (B_{ij})_{i,j\in E}$ be an integral matrix satisfying

$$B_{ij} = -B_{ji}, \qquad \det B = 1.$$

An entry $B_{ij} > 0$ denotes arrows from i to j, and the directed graph induced on M is assumed acyclic. Multiple arrows are allowed. The determinant hypothesis concerns the *whole* matrix, including the frozen rows and columns. The empty index set is allowed, with the usual conventions for empty products and the determinant of the empty matrix.

For $i \in M$, define the exchange monomials

$$P_i(a) = \prod_{j\in E} a_j^{\max(B_{ij},0)}, \qquad Q_i(a) = \prod_{j\in E} a_j^{\max(-B_{ij},0)}.$$

Our space is the actual coordinate locus

$$Y = \left\{(a,b) \in \mathbb{C}^E \times \mathbb{C}^M : a_i b_i = P_i(a) + Q_i(a)\ (i \in M), \quad a_f \neq 0\ (f \in F)\right\}, \tag{1}$$

with its inherited topology. In particular, the frozen coordinates are variables, rather than prescribed coefficients. Partner coordinates b_i occur only at mutable indices. The point $a_j = 1$, $b_i = 2$ shows that the space is nonempty.

The initial torus is

$$T = \{p \in Y : a_j(p) \neq 0 \text{ for all } j \in E\}.$$

Eliminating b_i identifies T with $(\mathbb{C}^*)^d$. On this torus, put

$$\Omega_T = \frac{1}{2} \sum_{i,j\in E} B_{ij} \frac{da_i}{a_i} \wedge \frac{da_j}{a_j}. \tag{2}$$

The factor $1/2$ compensates exactly once for the two ordered pairs (i,j) and (j,i).

Independently of any differential form or flow, define the coordinate cap

$$C = \left\{p \in Y : |a_f| = 1 \ (f \in F), \quad |a_i| \le 1, \quad (1-|a_i|^2)b_i = 0 \ (i \in M)\right\}. \tag{3}$$

Thus a partner must vanish when its original coordinate is strictly inside its unit disk; no additional partner condition is imposed by (3) when $|a_i| = 1$. The exchange equations always remain in force.

For a complete flow $\varphi : \mathbb{R} \times Y \to Y$, we use the following definition:

$$\mathrm{Skel}(\varphi) = \left\{p \in Y : \overline{\{\varphi_t(p) : t \ge 0\}} \text{ is compact in } Y\right\}. \tag{4}$$

This definition uses the entire positive orbit and the topology of Y. It does not assume a compact-domain presentation of the manifold.

Theorem 1.1. *The space Y admits a smooth real manifold structure of dimension $2d$ compatible with its inherited topology, for which the coordinate inclusion in $\mathbb{C}^E \times \mathbb{C}^M$ is a smooth immersion. The initial torus is dense. There is a smooth real one-form λ such that:*

1. *$\omega = d\lambda$ is symplectic and $\omega|_T = \operatorname{Im}\Omega_T$;*
2. *the vector field Z determined by $\iota_Z\omega = \lambda$ has a smooth flow φ defined for all real times;*
3. *for this same flow, $\mathrm{Skel}(\varphi) = C$.*

The statement proved here is about smooth real forms and complete Liouville dynamics. A holomorphic extension theorem is not a premise of the construction. The proof also does not need a Weinstein structure, finite-type conclusion, cap product decomposition, or categorical identification.

2 Integral node charts on the full coordinate locus

A subset $I \subseteq M$ is *independent* if $B_{ij} = 0$ for all $i, j \in I$. For a point $p \in Y$, write

$$I(p) = \{i \in M : a_i(p) = 0\}.$$

Lemma 2.1. *The set $I(p)$ is independent for every $p \in Y$.*

Proof. Suppose that the induced directed graph on $I(p)$ contains an edge. Choose a source in a nontrivial connected component of that graph; this is possible because the mutable graph is finite and acyclic. Let i be this source. At least one outgoing neighbor has zero coordinate, so $P_i(p) = 0$. There is no incoming neighbor in $I(p)$, and every frozen coordinate is nonzero; hence every factor in $Q_i(p)$ is nonzero. Thus $Q_i(p) \neq 0$. The equation $a_i b_i = P_i + Q_i$ becomes $0 = Q_i(p)$, a contradiction. Skew-symmetry then excludes both possible signs of an entry between two zero indices. □

For an independent I, let $L = E \setminus I$ and set

$$U_I = \{p \in Y : a_j(p) \neq 0 \text{ for all } j \in L\}.$$

For $i \in I$, every variable appearing in P_i or Q_i belongs to L. Consequently Q_i is a unit on U_I, and the expressions

$$q_i = P_i/Q_i = \prod_{j \in L} a_j^{B_{ij}}, \qquad v_i = b_i/Q_i, \qquad u_i = a_i \tag{5}$$

are well-defined there. Their exchange equation is

$$u_i v_i = 1 + q_i, \qquad q_i \neq 0. \tag{6}$$

Lemma 2.2. *The rows $(B_{ij})_{j\in L}$, indexed by $i \in I$, extend to an integral basis of $\mathbb{Z}^L$.*

Proof. Since $\det B = 1$, all rows of B form an integral basis of $\mathbb{Z}^E$. The span of the selected rows is therefore a direct summand, and in particular is saturated. These rows have zero I-coordinates by independence. Their span lies in the direct summand $\mathbb{Z}^L \subseteq \mathbb{Z}^E$ and remains saturated there: if a nonzero integer multiple of $x \in \mathbb{Z}^L$ belongs to the span, saturation in $\mathbb{Z}^E$ already puts x in that span. The selected rows are independent. Their quotient in $\mathbb{Z}^L$ is thus a finitely generated torsion-free abelian group, hence free. Lifting a basis of the quotient gives the required completion. □

Fix such a basis completion. It gives Laurent coordinates

$$z = (q_i \ (i \in I), w_1, \ldots, w_{d-2|I|})$$

on $(\mathbb{C}^*)^L$. The inverse change of variables also consists of Laurent monomials with integral exponents. Let F denote the matrix of the forward change and $D = F^{-1}$ its integral inverse, so that

$$a_j = \prod_k z_k^{D_{jk}} \qquad (j \in L).$$

There are $d-2|I|$ complementary coordinates because the selected independent rows number $|I|$ inside a lattice of rank $d - |I|$.

Eliminate q_i by $q_i = u_i v_i - 1$. For mutable $j \notin I$, eliminate the partner using $b_j = (P_j + Q_j)/a_j$. This produces an explicit coordinate isomorphism

$$U_I \simeq \{(u_i, v_i)_{i\in I} : u_i v_i \neq 1\} \times (\mathbb{C}^*)^{d-2|I|}. \tag{7}$$

The remaining partners are recovered by $b_i = Q_i v_i$. Every denominator used here is nonzero on the stated domain. In particular, this is an integral Laurent change of coordinates, with no finite covering.

The right side of (7) is an open subset of a complex vector space of dimension d, viewed as a real vector space of dimension $2d$. The forward and inverse maps are continuous in the inherited topology and smooth on their domains. On overlaps they give smooth transition maps. Lemma 2.1 ensures that a point belongs to $U_{I(p)}$, so these charts cover Y. They define the claimed real smooth structure. The coordinate inclusion is smooth; it is an immersion because the smooth chart map, defined using the ambient coordinates on a neighborhood, is a local left inverse of the coordinate inclusion and therefore makes its derivative injective.

In a node chart, T is obtained by also requiring $u_i \neq 0$ for each $i \in I$. The complement is a finite union of coordinate hyperplanes inside an open domain. The torus is therefore dense in every node chart, and hence dense in Y. We will use this density to glue actual smooth forms and to extend identities of continuous tensors.

3 The global real symplectic form

We retain the entire residual block in the node expression of (2). This avoids imposing a nondegeneracy assumption on an auxiliary form.

Let B_{LL} be the outside block of B, and put

$$R = D^{\mathsf{T}} B_{LL} D.$$

The indices of R are the q-indices and the complementary w-indices. On the node domain, define

$$\eta_i = \frac{d(u_i v_i - 1)}{u_i v_i - 1} \quad (i \in I), \qquad \eta_k = \frac{dw_k}{w_k} \quad (k \text{ complementary}).$$

Consider the complex-valued expression

$$\Omega_I = \sum_{i \in I} \frac{du_i \wedge dv_i}{u_i v_i - 1} + \frac{1}{2} \sum_{k,l} R_{kl} \eta_k \wedge \eta_l. \tag{8}$$

All its coefficients are smooth on the node domain. On the initial torus, independence splits (2) as

$$\Omega_T = \sum_{i \in I} d\log a_i \wedge d\log q_i + \frac{1}{2} \sum_{j,k \in L} B_{jk}\, d\log a_j \wedge d\log a_k.$$

The first summand has the exact cancellation

$$d\log u_i \wedge d\log(u_i v_i - 1) = \frac{du_i \wedge dv_i}{u_i v_i - 1}. \tag{9}$$

The inverse Laurent differential expresses the second summand by R. Thus (8) agrees with (2) on $T \cap U_I$.

Pull back $\operatorname{Im} \Omega_I$ by the actual node-chart differential. On overlaps the resulting real two-forms agree on the dense torus. Their coefficients are continuous, so they agree on the whole overlap. They glue to a smooth real two-form ω on Y. The identity $d\omega = 0$ follows in the same way: it holds on T by the logarithmic formula, and $d\omega$ is a smooth, hence continuous, tensor. This establishes a global real extension without using a separately assumed holomorphic extension.

Lemma 3.1. *The form ω is nondegenerate at every point of Y.*

Proof. Choose the chart indexed by the actual zero set $I = I(p)$. At its center $u_i = 0$ for all $i \in I$, so $q_i = -1$ and

$$\eta_i = -v_i\, du_i.$$

After the integral outside change of basis, the matrix of the original logarithmic form, in the order (a_I, q_I, w), is

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & R_{qq} & R_{qw} \\ 0 & R_{wq} & R_{ww} \end{pmatrix}. \tag{10}$$

It is invertible because it is congruent to the full invertible matrix B. In particular, R_{ww} is invertible. Indeed, if $R_{ww}x = 0$, then $(R_{qw}x, 0, x)$ lies in the kernel of (10), and hence $x = 0$. This proof also covers a zero-dimensional complementary block.

Now let a complex tangent vector at the center be in the kernel of Ω_I. Test it first against vectors supported in the v_i directions. The first term of (8) is $-\sum_i du_i \wedge dv_i$ there, while the residual term has no dv component. Thus the tested vector has zero u-components. Testing next against the free w-directions and using the invertibility of R_{ww} forces its w-components to vanish. Finally testing against the u_i-directions forces its v-components to vanish. Therefore the complex bilinear expression is nondegenerate at the center.

For a complex bilinear alternating form, nondegeneracy implies nondegeneracy of its imaginary part as a real form: if a real tangent vector x satisfies $\operatorname{Im}\Omega_I(x,y)=0$ for every y, testing also against iy gives $\operatorname{Re}\Omega_I(x,y)=0$. Hence $x=0$. The chart differential is a real tangent isomorphism, so the same conclusion holds for ω_p. Every point is the center of the chart indexed by its own zero set; no density argument for nondegeneracy is required. □

4 An explicit primitive on the torus

On T, write

$$A_j=\log|a_j|,\quad \theta_j=\arg(a_j),\quad S_i=\sum_j B_{ij}A_j=\log|q_i|,\quad T_i=\arg(-q_i).$$

Angles are circle-valued. Their differentials are globally defined on the torus, and a periodic scalar function of T_i is independent of its representative. Set

$$\beta=\sum_{i,j}A_iB_{ij}\,d\theta_j. \tag{11}$$

Expanding $d\log a_i=dA_i+\mathrm{i}\,d\theta_i$ and using skew-symmetry gives $d\beta=\operatorname{Im}\Omega_T$.

We give concrete cutoff choices. Write

$$e(x)=\begin{cases}e^{-1/x}, & x>0,\\ 0, & x\le 0,\end{cases}\qquad \sigma(x)=\frac{e(x)}{e(x)+e(1-x)}.$$

This is the smooth monotone transition used in Mathlib: $\sigma(x)=0$ for $x\le 0$, $\sigma(x)=1$ for $x\ge 1$, and $\sigma(1-x)=1-\sigma(x)$. Fix

$$\varepsilon>0,\qquad \kappa\ge 2\varepsilon+1,\qquad \delta>0. \tag{12}$$

The choice $(\varepsilon,\kappa,\delta)=(1,3,1)$ will suffice for the existence theorem.

Define

$$\chi(x)=\sigma\left(\frac{-2x-\varepsilon}{\varepsilon}\right),\qquad \rho(s)=\sigma(2-s^2), \tag{13}$$

$$\chi_i(A)=\prod_{\substack{j\in M\\ B_{ij}\neq 0}}\chi(A_i-A_j). \tag{14}$$

Then $\chi=1$ on $(-\infty,-\varepsilon]$ and $\chi=0$ on $[-\varepsilon/2,\infty)$. The function ρ equals one for $|s|\le 1$ and has compact support. Only mutable neighbors enter (14). In particular, frozen coefficients are not fixed or compared by this cutoff.

For adjacent mutable indices i,j, χ_i and χ_j cannot both be nonzero. Indeed nonvanishing would require both $A_i-A_j<-\varepsilon/2$ and $A_j-A_i<-\varepsilon/2$. This exclusion is the reason for using the dominance factors: a bounded ratio q_i alone does not detect whether several of its factors are approaching zero.

Let

$$L=2(\kappa-2\varepsilon)>0,\qquad G(x)=\int_0^x\sigma(t)\,dt,\qquad h(A)=-L\,G\left(\frac{-A-2\varepsilon}{L}\right).$$

The reflection identity for σ gives $G(1) = 1/2$, and direct differentiation yields

$$h \le 0, \qquad 0 \le h' \le 1, \qquad |h(A)| \le |A|, \tag{15}$$

$$h(A) = 0 \quad (A \ge -2\varepsilon), \qquad h(A) = A + \kappa \quad (A \le -L - 2\varepsilon), \tag{16}$$

$$A < h(A) \quad (A < 0). \tag{17}$$

For the last inequality, if $A < -2\varepsilon$, use $0 \le G(x) \le x$ for $x \ge 0$ to obtain $h(A) \ge A + 2\varepsilon > A$; in the other negative range it is immediate. All positive-order derivatives of h are bounded. For order at least two this also follows from compact support of the second derivative and its successive derivatives.

For the angular cutoff put $r = \min(\delta/2, \pi/4)$ and

$$W(T) = \sigma\left(\frac{\cos T - \cos r}{\cos(r/2) - \cos r}\right), \qquad \vartheta(T) = \arcsin(\sin T)\, W(T). \tag{18}$$

The denominator is positive. This is a smooth 2π-periodic function: the support of W avoids the nonsmooth points of $\arcsin(\sin T)$. On the principal arc, it equals T for $|T| \le r/2$ and is zero for $|T| \ge r$. In addition,

$$\vartheta'(T) \le 1, \qquad \sin T\, \vartheta(T) \ge 0, \qquad |\vartheta(T)| \le \frac{\pi}{2} |\sin T|. \tag{19}$$

For the derivative bound, on the support write $\vartheta' = W + TW'$. The second term is nonpositive, since W decreases away from zero on the principal arc, and $W \le 1$. The global sign and absolute-value estimates follow from the corresponding properties of arcsin and $0 \le W \le 1$. Periodicity gives uniform bounds on all derivatives.

We now define the torus Hamiltonian correction and primitive:

$$H = \sum_{i \in M} h(A_i)\chi_i(A)\rho(S_i)\vartheta(T_i), \qquad \lambda_T = \beta - dH. \tag{20}$$

Both are single-valued despite the angular notation. The identity $d\lambda_T = \operatorname{Im} \Omega_T$ is immediate. The work is to extend the one-form, since neither β nor H separately extends across the coordinate divisors.

5 Simultaneous cancellation at the boundary

Fix $p \in Y$ and $I = I(p)$. Work in U_I with $u_i = a_i$, $v_i = b_i/Q_i$. Shrink to an open neighborhood of p on which, for every $i \in I$, the following conditions hold on its torus part:

1. $\chi_i = 1$, while every correction centered at a mutable neighbor of I is identically zero;
2. $A_i < -L - 2\varepsilon$, so $h(A_i) = A_i + \kappa$;
3. the principal branch $\mathcal{L}_i = \operatorname{Log}(1 - u_i v_i)$ is defined and small enough that $\rho(\operatorname{Re} \mathcal{L}_i) = 1$ and $\vartheta(\operatorname{Im} \mathcal{L}_i) = \operatorname{Im} \mathcal{L}_i$.

These conditions can be imposed simultaneously. Every neighbor of a zero coordinate is nonzero by independence, so the required strict relative-norm inequalities hold near p. Also $u_i v_i = 0$ at p, and hence the logarithmic branch is defined near its value 1. The first condition makes neighboring corrections vanish on an open neighborhood, including all their derivatives.

On the torus part, $\mathcal{L}_i = S_i + \mathrm{i}T_i$, since $-q_i = 1 - u_i v_i$. The terms of β involving an index in I are exactly

$$\sum_{i \in I} (A_i \, dT_i - S_i \, d\theta_i).$$

There is no block with two indices in I. For each such index,

$$\begin{aligned} A_i \, dT_i - S_i \, d\theta_i - d((A_i + \kappa)T_i) &= -T_i \, dA_i - S_i \, d\theta_i - \kappa \, dT_i \\ &= -\operatorname{Im}\left(\mathcal{L}_i \frac{du_i}{u_i} + \kappa \, d\mathcal{L}_i \right). \end{aligned} \tag{21}$$

The potentially singular coefficient is removable. Namely,

$$\mathcal{L}_i / u_i = v_i F(u_i v_i), \qquad F(w) = \begin{cases} \operatorname{Log}(1-w)/w, & w \neq 0, \\ -1, & w = 0. \end{cases}$$

The power series $F(w) = -\sum_{n \geq 0} w^n/(n+1)$ near zero shows that F is smooth there. Thus (21) extends as an actual smooth real covector, even if both u_i and v_i vanish. At $u_i = 0$ its value is

$$(1 + \kappa) \operatorname{Im}(v_i \, du_i). \tag{22}$$

All outside terms of β involve units. Any correction not indexed by I or a neighbor of I depends only on nonzero coordinates near p, and is already smooth. Adding those regular terms to the extensions (21) defines a smooth local one-form. It agrees with λ_T on the dense torus. The local one-forms therefore agree on overlaps and glue to a global λ. Finally $d\lambda = \omega$ follows from equality on T and continuity of the two smooth two-forms. Together with Lemma 3.1, this gives an exact symplectic structure on the actual manifold Y.

6 The actual Liouville vector field and its boundary equations

Nondegeneracy determines a unique smooth real vector field Z by $\iota_Z \omega = \lambda$. Its smoothness follows locally by smoothly inverting the matrix of the nondegenerate two-form. We analyze this field, rather than prescribing a separate model flow.

On the torus set $f_i(A) = h(A_i)\chi_i(A)\rho(S_i)$. Contracting $\omega = \sum_{ij} B_{ij} \, dA_i \wedge d\theta_j$ gives

$$ZA_i = A_i - h(A_i)\chi_i(A)\rho(S_i)\vartheta'(T_i) \qquad (i \in M), \tag{23}$$

$$ZA_f = A_f \qquad (f \in F), \tag{24}$$

$$BZ\theta = \partial_A H. \tag{25}$$

There are no derivatives of χ_i in (23); such derivatives occur in the angular equation instead. The inverse in (25) is the inverse of the full matrix B.

Since $0 \leq \chi_i, \rho \leq 1$ and $\vartheta' \leq 1$, the product $\chi_i \rho \vartheta'$ is at most one, even where ϑ' is negative. Equations (15)–(17) imply

$$\begin{aligned} A_i \geq -2\varepsilon \quad &\Longrightarrow ZA_i = A_i, \\ A_i < 0 \quad &\Longrightarrow ZA_i \leq A_i - h(A_i) < 0. \end{aligned} \tag{26}$$

There is also a constant $c > 0$, depending only on the fixed seed and cutoffs, such that

$$|ZA_j| \leq c|A_j|. \tag{27}$$

For a mutable index this follows from $|h(A_j)| \le |A_j|$ and a uniform bound for $|\vartheta'|$; for a frozen index it follows from (24).

Each of these statements extends to the entire open locus $a_j \neq 0$, even when other coordinates vanish. On that locus both $A_j = \log|a_j|$ and $ZA_j = \mathrm{Re}(Za_j/a_j)$ are smooth. The equality or weak inequality holds on its dense intersection with T and hence everywhere. The strict negative conclusion follows from the already strict scalar inequality $A_j - h(A_j) < 0$.

To control zero coordinates, return to the localized neighborhood of Section 5. There is a smooth real function ν_i on this neighborhood for which

$$Zu_i = (-\kappa + \mathrm{i}\nu_i)u_i, \qquad Z(u_iv_i) = (u_iv_i - 1)\,\mathrm{Log}(1 - u_iv_i). \tag{28}$$

Here is why the angular coefficient is genuinely smooth. In $BZ\theta = \partial_A H$, the singular corrections have the localized form $(A_i + \kappa)T_i$. Their radial derivative is T_i, with no remaining logarithmic divergence; neighboring corrections vanish identically. The other derivatives are those of smooth functions of the nonzero coordinates. Multiplication by the constant matrix B^{-1} gives a smooth real angular vector. On the dense torus its ith component is the coefficient ν_i above. Moreover every neighbor of $i \in I$ has uncorrected radial drift in this neighborhood, so $ZS_i = S_i$, while $ZT_i = \partial_{A_i} H = T_i$. Differentiating $u_iv_i = 1 + q_i$ then proves the second equation. Smoothness extends both equations across $u_i = 0$.

Define near zero

$$E(w) = \begin{cases} (w-1)\,\mathrm{Log}(1-w)/w, & w \neq 0, \\ 1, & w = 0. \end{cases}$$

Differentiating u_iv_i in (28) first where $u_i \neq 0$ and then using continuity gives

$$Zv_i = (E(u_iv_i) + \kappa - \mathrm{i}\nu_i)v_i.$$

Consequently the exact equation on the zero divisor is

$$u_i = 0 \quad \Longrightarrow \quad Zv_i = (1 + \kappa - \mathrm{i}\nu_i)v_i. \tag{29}$$

No division by a zero original coordinate or a possibly zero partner is used in the final identity.

Lemma 6.1. *Along an integral curve on a connected time interval, the set of vanishing original coordinates is constant. For an index in that set,*

$$|v_i(t)|^2 = e^{2(1+\kappa)(t-s)}|v_i(s)|^2. \tag{30}$$

Proof. Near any time at which $a_i = 0$, (28) gives the scalar equation

$$\frac{d}{dt}|a_i|^2 = -2\kappa|a_i|^2.$$

The integrating factor $e^{2\kappa t}$ shows that a zero value persists throughout a small connected time neighborhood, in both directions. Thus the zero-time set is open in the curve's time interval; it is also closed by continuity. Connectedness shows that it is empty or the whole interval. Frozen coordinates are never zero by definition of Y. This proves constancy of the whole zero set, without assuming completeness beforehand.

For i in that set, Q_i is nonzero at every curve point by Lemma 2.1, so $v_i = b_i/Q_i$ is smooth along the curve. Taking the real part of (29) gives $(|v_i|^2)' = 2(1+\kappa)|v_i|^2$, and the integrating-factor identity gives (30). This includes $v_i(s) = 0$. □

7 Finite-time confinement and the complete flow

Lemma 7.1. *Fix $p \in Y$ and $T_0 \geq 0$. There is a compact subset $K(p, T_0)$ of Y such that any integral curve of Z starting at p remains in $K(p, T_0)$ at every time in its interval of definition with $|t| \leq T_0$.*

Proof. Let $I = I(p)$. By Lemma 6.1, precisely the coordinates indexed by I vanish along such a curve. For $j \notin I$, apply Gronwall's inequality to (27), in both time directions, to obtain

$$|A_j(t)| \leq |A_j(0)|e^{c|t|}.$$

Let $R = \max_{j \notin I} |A_j(0)|e^{cT_0}$, with $R = 0$ if the index set is empty. Then every nonzero original coordinate satisfies

$$e^{-R} \leq |a_j(t)| \leq e^R \qquad (|t| \leq T_0). \tag{31}$$

The remaining originals are identically zero.

If a mutable i is not in I, its partner is $b_i = (P_i + Q_i)/a_i$. Each exchange monomial is a finite product of nonnegative powers of the bounded originals, and the denominator is bounded below by (31); thus b_i is uniformly bounded on the specified finite time range. If $i \in I$, use $b_i = Q_i v_i$. The monomial Q_i involves only nonzero coordinates and is uniformly bounded by (31), while (30) gives $|v_i(t)| \leq e^{(1+\kappa)T_0}|v_i(0)|$. Hence these partners are also uniformly bounded. This split between the two types of partner is necessary; one cannot use $b_i = (P_i + Q_i)/a_i$ at $i \in I$.

Choose common finite bounds A_0, B_0 for all originals and partners, and put $\eta = e^{-R} > 0$. The coordinate box

$$\{q \in Y : |a_j(q)| \leq A_0,\ |b_i(q)| \leq B_0,\ |a_f(q)| \geq \eta\ (f \in F)\}$$

contains every such curve segment. Its image in $\mathbb{C}^E \times \mathbb{C}^M$ is closed and bounded: the exchange equations and the displayed inequalities are closed conditions, and the positive frozen lower bound implies all the nonvanishing conditions defining Y. Finite-dimensional compactness therefore proves the claim in the inherited topology. □

Proposition 7.2. *The vector field Z has a unique smooth complete flow $\varphi : \mathbb{R} \times Y \to Y$.*

Proof. A smooth vector field has local solutions, uniqueness, and smooth dependence in the node charts. On a compact set, finitely many local existence neighborhoods yield a common positive lifetime for solutions starting at any point of that set.

Suppose a maximal solution through p had a finite right endpoint b. Lemma 7.1, applied to a time bound larger than $|b|$, puts its segment approaching b in a fixed compact set. Let $\tau > 0$ be a common two-sided lifetime for initial points of that set. Start a local solution at a time $s < b$ with $b - s < \tau/2$. Uniqueness identifies it with the old solution on their overlap, and it extends the old solution beyond b, a contradiction. The same argument at a finite left endpoint proves existence for all real times.

Set $\varphi_t(p)$ equal to this all-real solution at time t. Uniqueness yields $\varphi_0 = \mathrm{id}$ and $\varphi_{s+t} = \varphi_s \circ \varphi_t$. Local smooth dependence, continued along a finite orbit segment by finitely many local flow maps, gives joint smoothness in (t, p). The resulting flow is the flow of the already constructed field Z, and its orbit derivative is $Z(\varphi_t(p))$. □

8 Compactness and positive invariance of the coordinate cap

Lemma 8.1. *The set C is compact. At a point of C, the set $J = \{i \in M : |a_i| < 1\}$ is independent, and every neighbor of an index in J has modulus one.*

Proof. All original coordinates have modulus at most one. If $|a_i| < 1$, the cap equation gives $b_i = 0$. If $|a_i| = 1$, the exchange equation gives $|b_i| \le |P_i| + |Q_i| \le 2$. The cap is thus bounded in the ambient coordinate space. Its image there is closed, being defined by the exchange equations, the frozen unit-modulus equations, and the closed conditions in (3). The frozen equations already imply nonvanishing. Hence it is compact.

For independence, if the mutable graph induced on J had an edge, choose a source in a nontrivial connected component. At that source i the partner vanishes, so $P_i + Q_i = 0$. Every incoming factor has modulus one, while at least one outgoing factor has modulus less than one. Thus $|Q_i| = 1$ and $|P_i| < 1$, a contradiction. Therefore J is independent. All mutable coordinates outside J have modulus one by definition, and all frozen coordinates have modulus one by the cap equations. This proves the neighbor assertion. □

We record a character equation that will be used in both inclusions. Suppose $a_i \neq 0$ and every full-row neighbor of i has modulus one. Then $Q_i \neq 0$, $|q_i| = 1$, and $S_i = 0$. Put

$$g_i(x) = h(x) \prod_{\substack{j \in M \\ B_{ij} \neq 0}} \chi(x), \qquad c_i(x) = g_i'(x).$$

In this formula the neighbors have logarithmic radius zero. The factor $\rho(S_i) = \rho(0) = 1$ has no derivative with respect to A_i, since $B_{ii} = 0$. The full angular equation gives the exact identity

$$\frac{d}{dt} q_i = \mathrm{i}\, q_i\, c_i(A_i)\, \vartheta(\arg(-q_i)). \tag{32}$$

Indeed terms centered at a mutable neighbor of i have both $h(0)$ and $h'(0)$ equal to zero; terms at nonneighbors have no A_i dependence. Only the term centered at i remains in $\partial_{A_i} H$. The same calculation is valid on simultaneous zero strata elsewhere: only row-support coordinates enter this character and they are nonzero.

There is a useful branch-free uniqueness estimate. From (19) and the polar sine identity,

$$|z|\,|\vartheta(\arg(-z))| \le \frac{\pi}{2} |\operatorname{Im} z| \le \frac{\pi}{2} |z + 1|. \tag{33}$$

Thus a solution of (32) that equals -1 at one time remains -1 on any finite time interval on which the coefficient is continuous. To see this directly, bound $|c_i(A_i)|$ on that interval and apply Gronwall to $|q_i + 1|$ using (33). Reversing the time parameter gives the same uniqueness in the backward direction. This argument does not differentiate a discontinuous choice of argument.

Proposition 8.2. *The cap is positively invariant under the complete flow φ. Consequently $C \subseteq$ Skel(φ).*

Proof. Let $p \in C$. First, any original coordinate initially of modulus one stays of modulus one: it is nonzero throughout the orbit by Lemma 6.1, and its logarithmic radius is identically zero by (27) and Gronwall. Every initially nonzero coordinate of modulus less than one stays strictly inside the disk, since (26) makes its negative logarithmic radius decrease. These statements use the vector field equations and the initial values; they do not assume cap membership at later times.

Consider an initially nonzero strict-interior mutable coordinate a_i. Lemma 8.1 gives unit modulus at all its neighbors, and the preceding zero-log argument propagates that condition along the orbit. The cap equation initially gives $b_i = 0$, so the normalized exchange equation gives $q_i = -1$. Equation (32) and the uniqueness estimate imply $q_i(t) = -1$ for every $t \ge 0$. Since $a_i(t)$ and $Q_i(t)$ remain nonzero, the exchange equation then gives $b_i(t) = 0$.

If $a_i(p) = 0$, it stays zero by Lemma 6.1. Its initial partner is zero by the cap condition, so (30) gives $v_i(t) = 0$ and hence $b_i(t) = 0$. At indices with initial modulus one, no partner restriction is needed because $1 - |a_i(t)|^2 = 0$. Frozen coordinates retain modulus one as well. All the conditions of (3) therefore hold for all positive times.

The entire forward orbit of p lies in the compact set C. Its closure is a closed subset of C, so is compact. This proves the stated inclusion using the independent definition (4). □

9 Every trapped orbit lies in the cap

Let $p \in \mathrm{Skel}(\varphi)$ and write $\gamma(t) = \varphi_t(p)$. Denote its compact forward-orbit closure by K. In particular, all original and partner coordinates are bounded on K, and every frozen modulus is bounded below by a positive constant on K.

9.1 Radial restrictions and independent limiting directions

For a frozen coordinate the exact solution of (24) is $A_f(t) = e^t A_f(0)$. A positive initial value makes $|a_f|$ unbounded, while a negative value makes it tend to zero, contradicting the positive lower bound on K. Thus $|a_f(p)| = 1$.

For any mutable coordinate with $A_i(0) > 0$, equation (26) gives the same positive exponential law and contradicts compactness. Hence every original modulus is at most one initially. For $0 < |a_i(p)| < 1$, its logarithmic radius remains negative and strictly decreases. It tends to $-\infty$: if it had a finite negative limit ℓ, the continuous strict bound $A - h(A) < 0$ would give $A_i' \le -c_0 < 0$ once A_i is sufficiently close to ℓ, contradicting convergence to ℓ. Consequently

$$|a_i(p)| < 1 \quad \Longrightarrow \quad |a_i(\gamma(t))| \longrightarrow 0, \tag{34}$$

including an initially zero coordinate, which stays zero.

Set $J = \{j \in E : |a_j(p)| < 1\}$. No frozen index lies in J. Choose a sequence of times tending to $+\infty$ and a convergent subsequence of their points in K. By (34), its limit has every J-coordinate zero. The zero set at the limit is independent by Lemma 2.1; therefore J is independent. Every neighbor of an index in J has initial modulus one and hence modulus one for the entire orbit, by the zero-log law. This compact-limit argument includes arbitrary simultaneous-zero strata; it does not restrict the orbit closure to a single open node chart.

9.2 Nonzero interior coordinates: angular rigidity

Fix a mutable i with $0 < |a_i(p)| < 1$. All its neighbors have modulus one, so $|Q_i(\gamma(t))| = 1$ and the genuine character $q_i = P_i/Q_i$ satisfies (32) throughout the orbit. Boundedness of b_i on K and $a_i(t) \to 0$ give

$$q_i(t) + 1 = \frac{a_i(t) b_i(t)}{Q_i(t)} \longrightarrow 0. \tag{35}$$

Eventually $A_i(t) < -L - 2\varepsilon$. In this deep region the neighbor radii are zero, so each dominance factor is one and $g_i(A_i) = A_i + \kappa$. Thus $c_i(A_i) = 1$ at all sufficiently large times.

To conclude rigidity without choosing angle lifts, differentiate the squared distance to the equilibrium:

$$\frac{d}{dt}|q_i + 1|^2 = 2c_i(A_i)\big(-\operatorname{Im} q_i\big)\vartheta(\arg(-q_i)). \tag{36}$$

The last two factors have nonnegative product, because $-\operatorname{Im} q_i = |q_i| \sin(\arg(-q_i))$ and $\sin T\,\vartheta(T) \geq 0$. Therefore $|q_i + 1|^2$ is nondecreasing for all sufficiently large times. By (35) it tends to zero, so it must already be identically zero on that tail.

Apply the finite-interval backward uniqueness following (33) to propagate $q_i = -1$ from the tail back to the initial time. This step is necessary: in the transition region the derivative $c_i(A_i)$ need not be nonnegative, and the dominance factor need not be one. We use only continuity of that coefficient on each finite interval. Finally $a_i(p) \neq 0$ and $Q_i(p) \neq 0$ in the exchange equation give $b_i(p) = 0$.

9.3 Initially zero coordinates: normalized-partner escape

Now let $a_i(p) = 0$. Its entire forward orbit lies in the closed zero divisor $D_i = \{a_i = 0\}$. On this whole divisor Q_i is nonzero: at each point, Lemma 2.1 says that every neighbor of i is nonzero. Thus b_i/Q_i is continuous at every point of D_i, including points with additional zero coordinates away from the neighbors of i.

The intersection $K \cap D_i$ is compact, so $|b_i/Q_i|$ is bounded there. Formula (30), however, gives

$$|b_i(\gamma(t))/Q_i(\gamma(t))|^2 = e^{2(1+\kappa)t}|b_i(p)/Q_i(p)|^2.$$

Since $1 + \kappa > 0$, boundedness forces the initial normalized partner to be zero, and hence $b_i(p) = 0$. It would not suffice here to assume that a normalization is bounded in one chosen open chart; compactness is applied on the entire closed zero slice.

Proposition 9.1. $\mathrm{Skel}(\varphi) \subseteq C$.

Proof. The radial argument gives $|a_f(p)| = 1$ for frozen indices and $|a_i(p)| \leq 1$ for mutable indices. If $|a_i(p)| < 1$, the two preceding cases give $b_i(p) = 0$, according as $a_i(p)$ is nonzero or zero. If $|a_i(p)| = 1$, the product $(1 - |a_i(p)|^2)b_i(p)$ vanishes automatically. These are exactly the conditions defining C. □

Propositions 8.2 and 9.1 establish the literal set equality for the same complete flow. Together with the manifold construction and $d\lambda = \omega$, this proves Theorem 1.1.

10 Correspondence with the formal statement

The point type in the formalization is `Seed.Point`: the coordinate subtype (1), with its inherited topology. `SmoothPresentation` requires the actual coordinate inclusion to be a smooth immersion. The real tangent spaces, manifold derivatives, exterior derivative, and flow are native Mathlib objects used through the pinned `FsFltz.Geometry` interfaces. The construction is therefore on this coordinate space, not on an unrelated carrier transported by an arbitrary point equivalence.

The final theorem is

`ClusterSkeleton.FullSkeleton.firstGeometricTarget.`

Its quantified seed hypotheses are precisely skew-symmetry, mutable acyclicity, and determinant one for the full exchange matrix. The parameterized construction accepts (12), and the unparameterized existence result chooses $(1, 3, 1)$. None of these three parameters fixes a frozen coordinate.

The two sides of the equality are defined independently: `coordinateCap` is (3), while `trappedSet` is compactness of the closure of the forward orbit of a Mathlib `Flow`. The final witnesses contain the global primitive, its Liouville vector field, and the constructed complete flow. Their contraction equation, all-real existence, joint smoothness, and the skeleton equality are proved properties; they are not supplied assumptions in the final existence theorem.

Mathematical step	Main source files or final declaration
Exchange equations and zero-set independence	`Seed.lean`, `ZeroSet.lean`
Integral node coordinates and native smooth atlas	`IndependentLaurentCoordinates.lean`, `IndependentNodeChart.lean`, `SmoothAtlas.lean`
Density and full local two-form	`InitialTorusDensity.lean`, `NodeTwoForm.lean`
Residual-block nondegeneracy and global symplectic form	`NodeNondegeneracy.lean`, `GlobalTwoForm.lean`, `GlobalSymplectic.lean`
Explicit cutoffs and simultaneous primitive cancellation	`PrimitiveCutoffs.lean`, `DominanceCutoff.lean`, `NodePrimitiveCancellation.lean`, `LocalPrimitive.lean`
Global primitive and its actual field	`GlobalPrimitive.lean`, `GlobalLiouvilleField.lean`
Zero-stratum invariance and complex partner dynamics	`ZeroStratumInvariance.lean`, `LocalAngularDynamics.lean`, `ZeroStratumPartnerDynamics.lean`
Finite-time compact confinement and all-real native flow	`FiniteTimeConfinement.lean`, `NativeCompleteFlow.lean`
Cap compactness, interior partner propagation, and invariance	`CapSet.lean`, `InteriorPartnerPropagation.lean`, `CapForwardInvariance.lean`
Compact-orbit rigidity and zero-partner escape	`CompactInteriorRigidity.lean`, `ZeroStratumEscape.lean`
The two inclusions for the constructed flow	`ConstructedFlowCapInclusion.lean`, `ConstructedFlowReverseInclusion.lean`
The literal equality and full existence theorem	`FullSkeleton.trappedSet_eq_coordinateCap`, `FullSkeleton.liouvilleSkeletonTarget`, `FullSkeleton.firstGeometricTarget`

The accompanying verification records concern this geometric theorem. The proof above uses no conclusion from homological mirror symmetry or from a finite-type or cocore construction. Its scope is exactly the real Liouville structure and full trapped-set equality stated in Theorem 1.1.

Sources. The motivating geometric argument is the Liouville and skeleton portion of the supplied *acyclic_cluster_HMS* notes, Section 2. This exposition has been rewritten to match the formalized real-geometric construction; it does not reproduce the unrelated parts of those notes. The Lean project uses Mathlib and Bohan Fang's `fs-fltz-minimal` as pinned dependencies. Their exact revisions and the original manuscript hash are recorded in `docs/sources.json`.